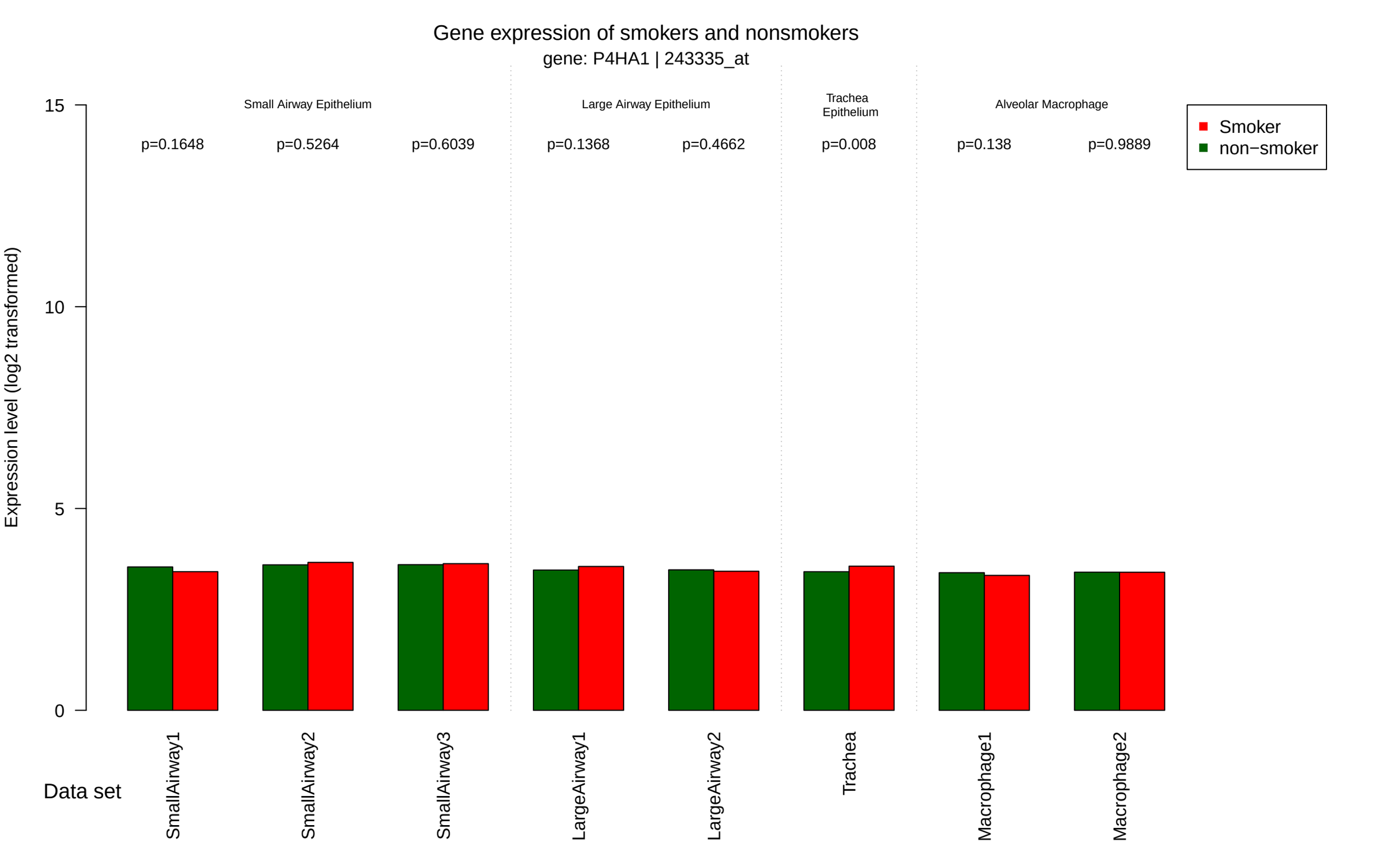
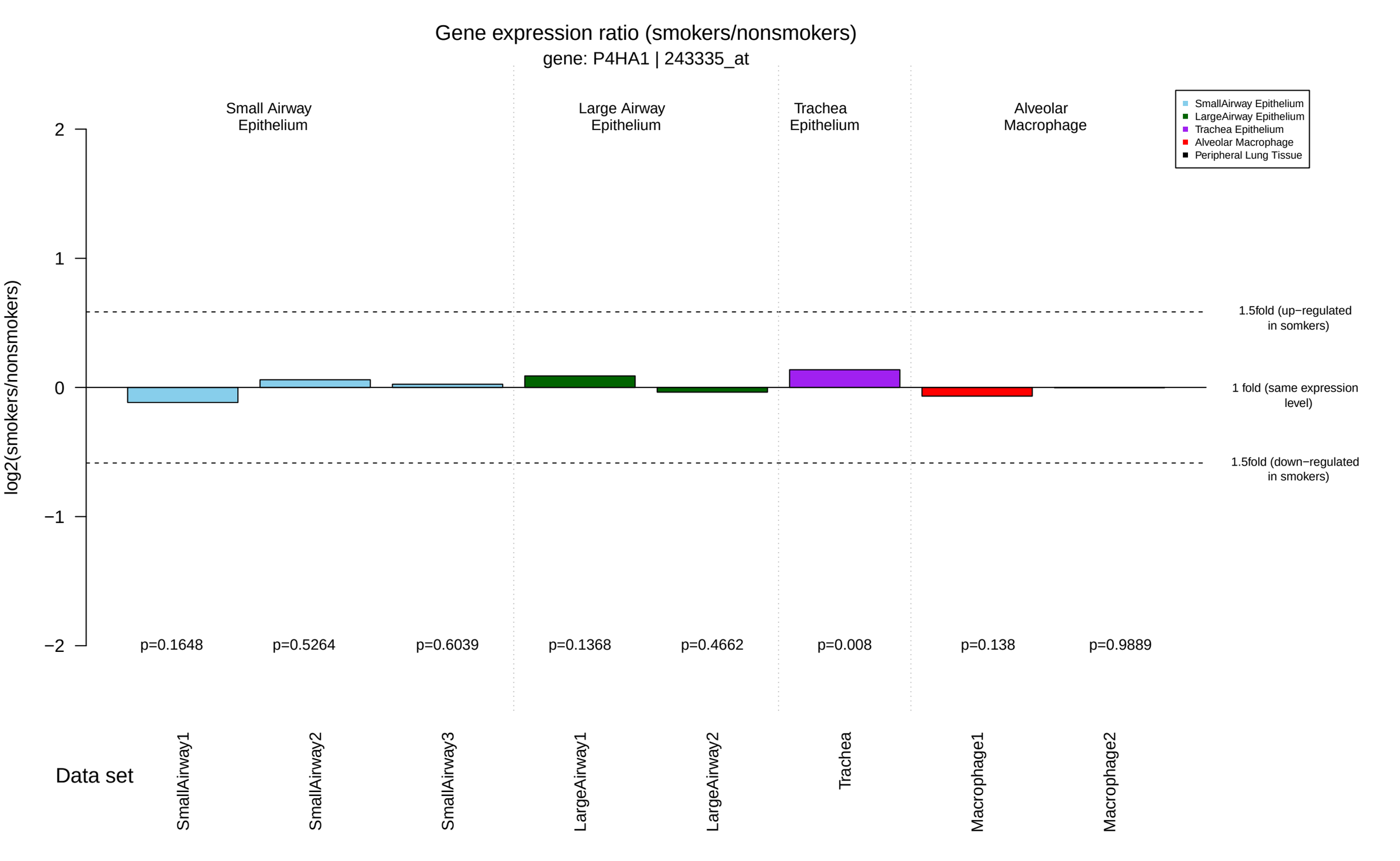
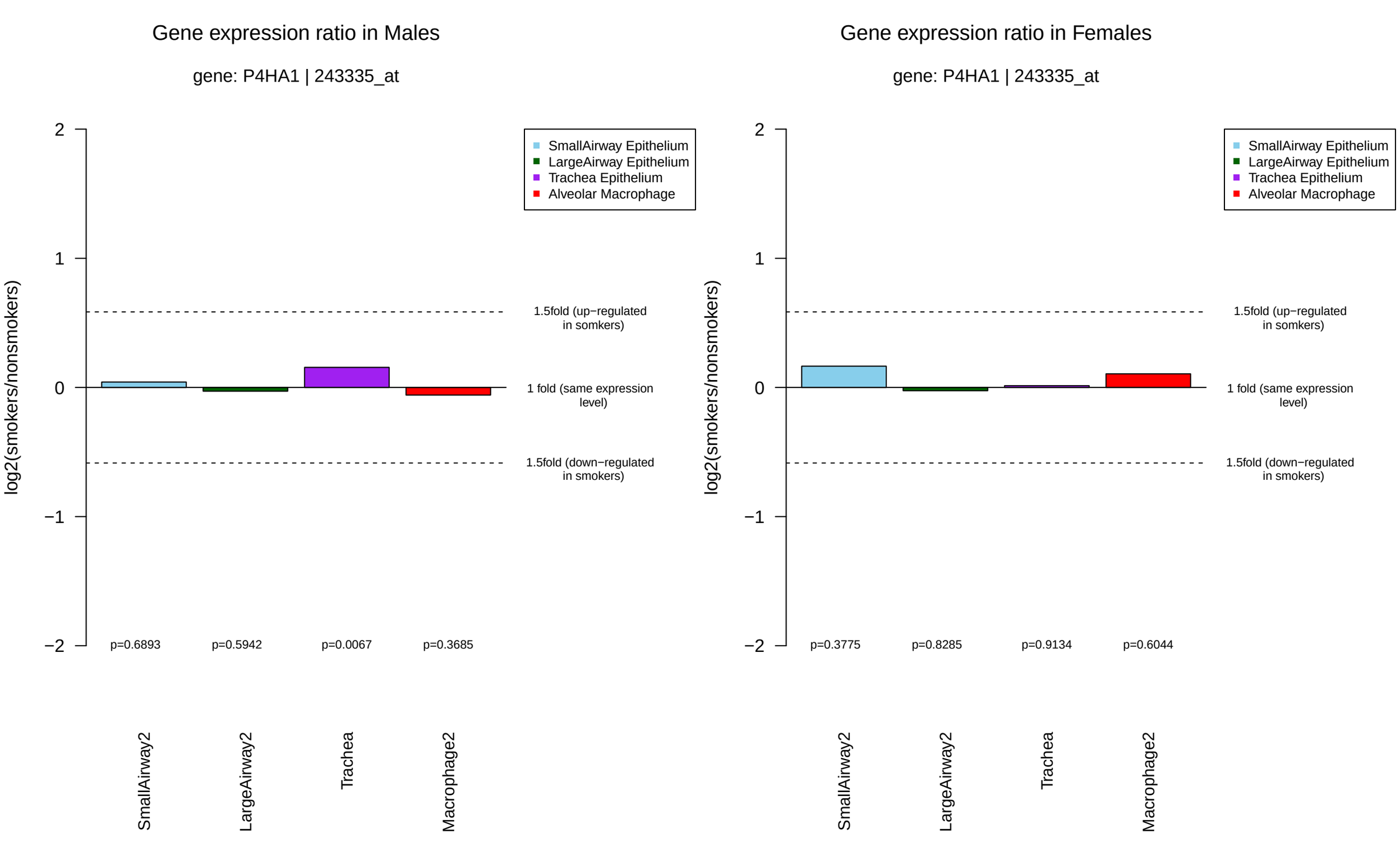
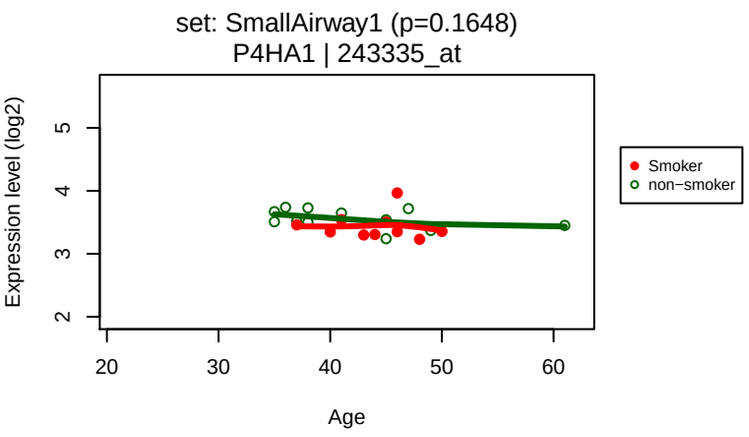
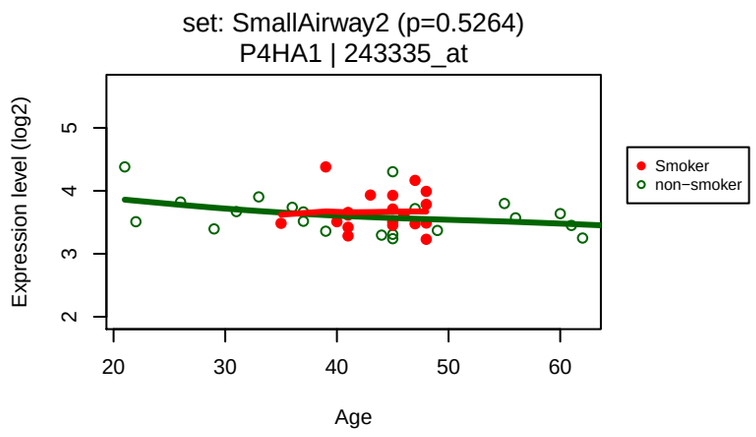
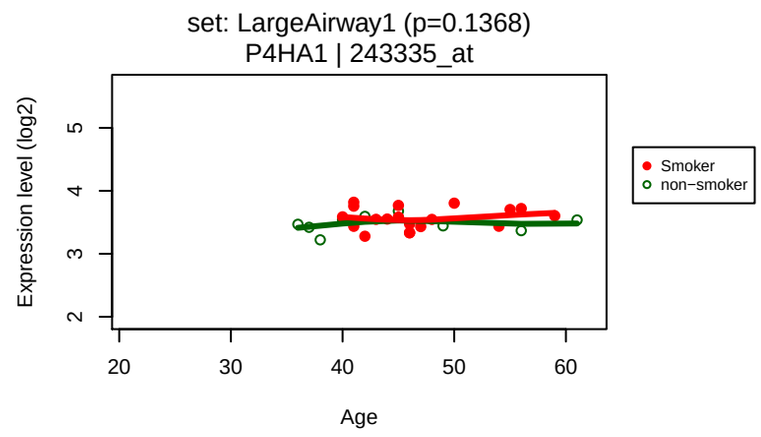

Gene expression of smokers and nonsmokers
gene: P4HA1 | 243335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1648
p=0.5264
p=0.6039
p=0.1368
p=0.4662
p=0.008
p=0.138
p=0.9889
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

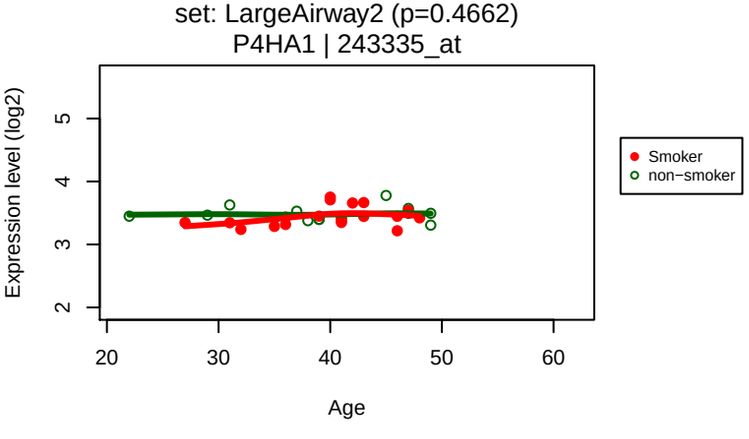

Gene expression ratio (smokers/nonsmokers)
gene: P4HA1 | 243335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1648
p=0.5264
p=0.6039
p=0.1368
p=0.4662
p=0.008
p=0.138
p=0.9889
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

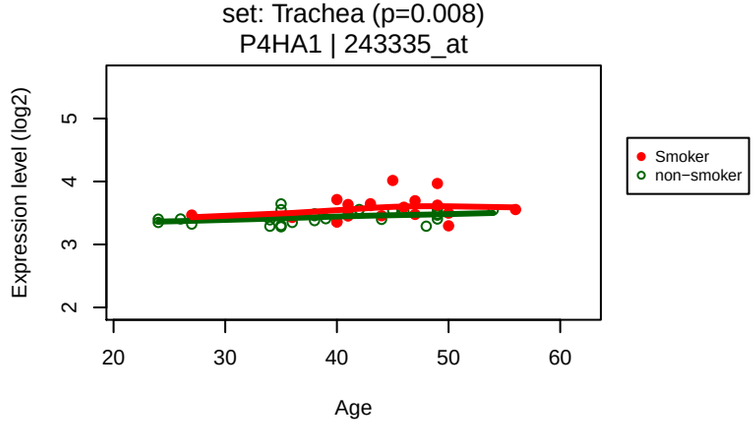

Gene expression ratio in Males
gene: P4HA1 | 243335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6893
p=0.5942
p=0.0067
p=0.3685
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: P4HA1 | 243335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3775
p=0.8285
p=0.9134
p=0.6044
SmallAirway2
LargeAirway2
Trachea
Macrophage2

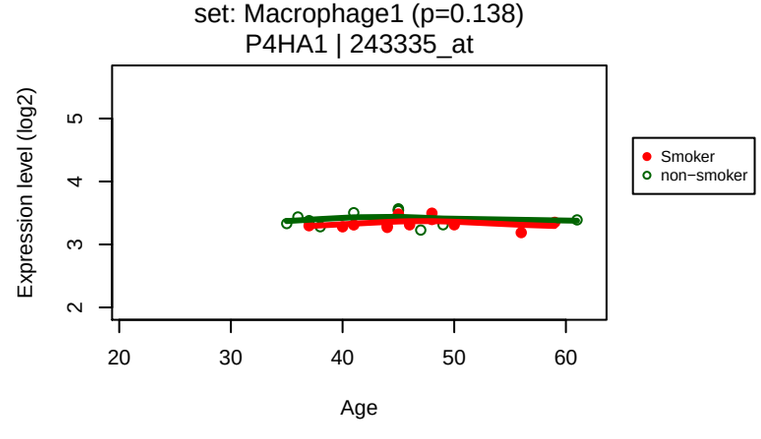

set: SmallAirway1 (p=0.1648)
P4HA1 | 243335_at
Expression level (log2)
Age
Smoker
non−smoker

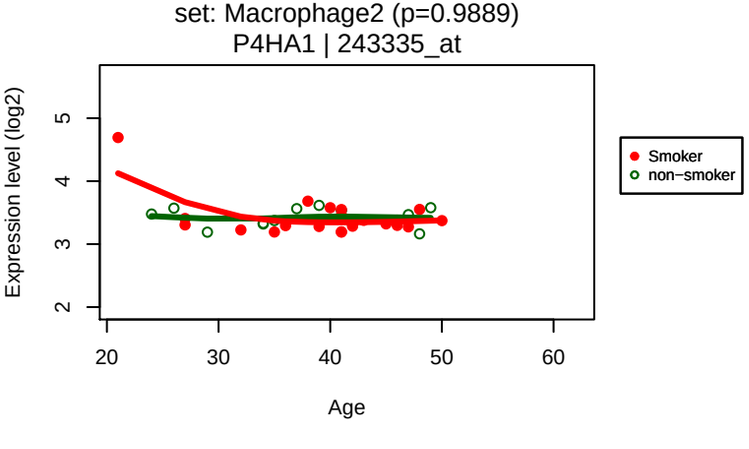

set: SmallAirway2 (p=0.5264)
P4HA1 | 243335_at
Expression level (log2)
Age
Smoker
non−smoker

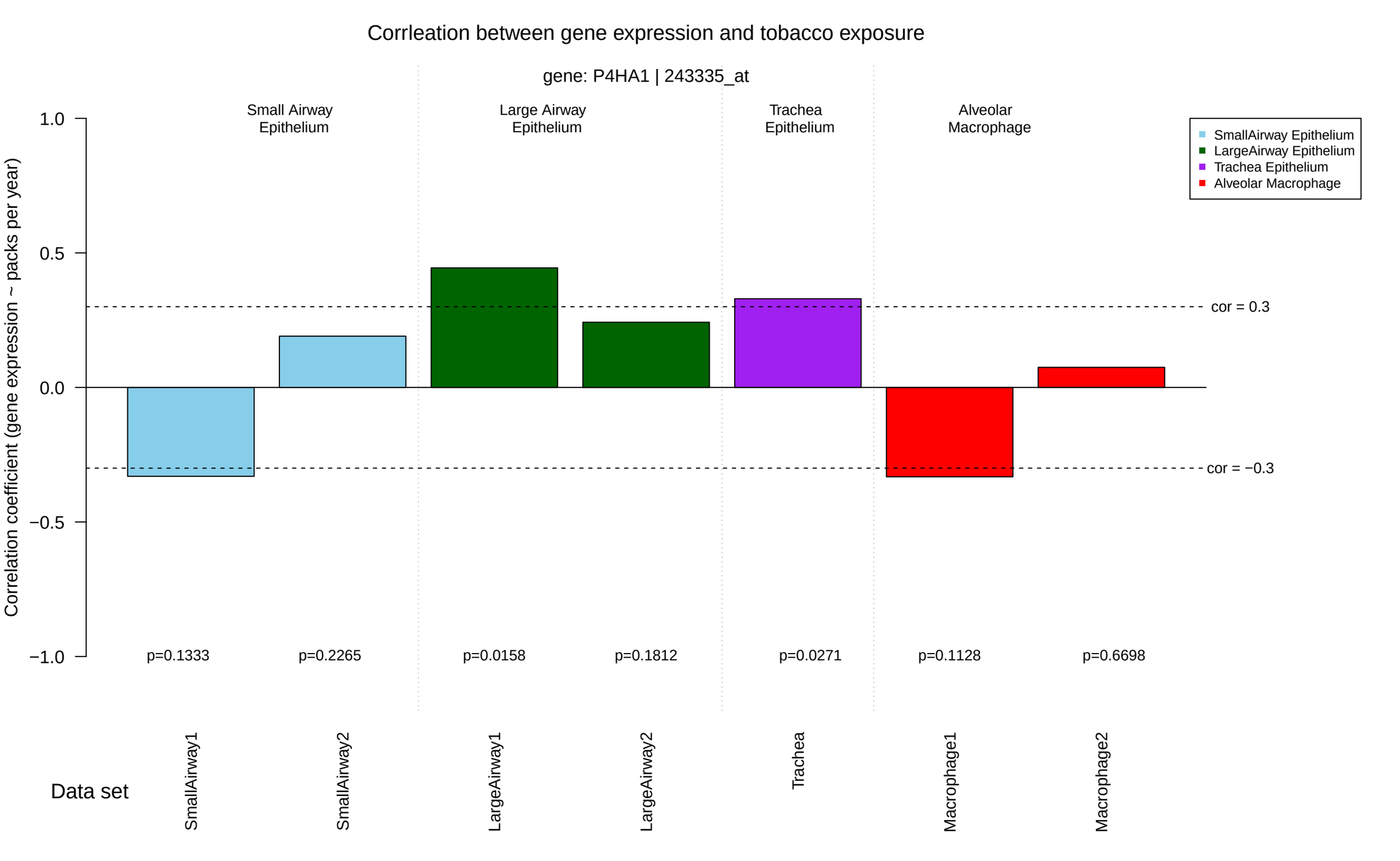

set: LargeAirway1 (p=0.1368)
P4HA1 | 243335_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4662)
P4HA1 | 243335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.008)
P4HA1 | 243335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.138)
P4HA1 | 243335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9889)
P4HA1 | 243335_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: P4HA1 | 243335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1333
p=0.2265
p=0.0158
p=0.1812
p=0.0271
p=0.1128
p=0.6698
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2